

- 4 By considering the binomial expansions of $\left(z + \frac{1}{z}\right)^5$ and $\left(z - \frac{1}{z}\right)^5$, where $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that

$$\tan^5 \theta = \frac{\sin 5\theta - a \sin 3\theta + b \sin \theta}{\cos 5\theta + a \cos 3\theta + b \cos \theta},$$

where a and b are integers to be determined.

[7]

[illegible]